Activity_main.xml

-

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Hello World!"
      app:layout_constraintBottom_toBottomOf="parent"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.simpledownstreammessaging;

import android.Manifest;
import android.content.pm.PackageManager;
import android.os.Build;
import android.os.Bundle;
import android.util.Log;

import androidx.activity.EdgeToEdge;
import androidx.activity.result.ActivityResultLauncher;
import androidx.activity.result.contract.ActivityResultContracts;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.content.ContextCompat;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import com.google.firebase.messaging.FirebaseMessaging;

public class MainActivity extends AppCompatActivity {

   private final ActivityResultLauncher<String> requestPermissionLauncher =
```

```
    registerForActivityResult(
            new ActivityResultContracts.RequestPermission(),
            isGranted -> {
                if (isGranted) {
                    Log.d("FCM", "Notification permission granted");
                } else {
                    Log.d("FCM", "Notification permission denied");
                }
            }
    );

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);

    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_main);

    // Ask for notification permission on Android 13+
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
        if (ContextCompat.checkSelfPermission(
                this,
                Manifest.permission.POST_NOTIFICATIONS
        ) != PackageManager.PERMISSION_GRANTED) {

            requestPermissionLauncher.launch(
                    Manifest.permission.POST_NOTIFICATIONS
            );
        }
    }

    // Get Firebase registration token
    FirebaseMessaging.getInstance().getToken()
            .addOnCompleteListener(task -> {
                if (!task.isSuccessful()) {
                    Log.e("FCM", "Failed to get token", task.getException());
                    return;
                }

                String token = task.getResult();
                Log.d("FCM_TOKEN", token);
            });

    ViewCompat.setOnApplyWindowInsetsListener(
            findViewById(R.id.main),
            (v, insets) -> {
                Insets systemBars =
                        insets.getInsets(WindowInsetsCompat.Type.systemBars());
```

```
            v.setPadding(
                systemBars.left,
                systemBars.top,
                systemBars.right,
                systemBars.bottom
            );

            return insets;
        });
    }
}
```

build.gradle.kts(SimpleDownstreamMessaging)

```
// Top-level build file where you can add configuration options common to all sub-projects/modules.
plugins {
    alias(libs.plugins.android.application) apply false
    id("com.google.gms.google-services") version "4.5.0" apply false
}
```

Build.gradle.kts(:app)

```
plugins {
    alias(libs.plugins.android.application)
    id("com.google.gms.google-services")
}

android {
    namespace = "com.example.simpledownstreammessaging"
    compileSdk {
        version = release(36) {
            minorApiLevel = 1
        }
    }

    defaultConfig {
        applicationId = "com.example.simpledownstreammessaging"
        minSdk = 24
        targetSdk = 36
        versionCode = 1
        versionName = "1.0"

        testInstrumentationRunner = "androidx.test.runner.AndroidJUnitRunner"
    }

    buildTypes {
        release {
```

```
        optimization {
            enable = false
        }
    }
  }
  compileOptions {
    sourceCompatibility = JavaVersion.VERSION_11
    targetCompatibility = JavaVersion.VERSION_11
  }
}

dependencies {
  implementation(platform("com.google.firebase:firebase-bom:34.18.0"))
  implementation("com.google.firebase:firebase-messaging")
  implementation(libs.activity.ktx)
  implementation(libs.appcompat)
  implementation(libs.constraintlayout)
  implementation(libs.material)
  testImplementation(libs.junit)
  androidTestImplementation(libs.espresso.core)
  androidTestImplementation(libs.ext.junit)
}
```

MyFirebaseMessagingService.java

```
package com.example.simpledownstreammessaging;

import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.os.Build;

import androidx.annotation.NonNull;
import androidx.core.app.NotificationCompat;

import com.google.firebase.messaging.FirebaseMessagingService;
import com.google.firebase.messaging.RemoteMessage;

public class MyFirebaseMessagingService extends FirebaseMessagingService {

  @Override
  public void onMessageReceived(@NonNull RemoteMessage remoteMessage) {

    String title = remoteMessage.getNotification().getTitle();
    String message = remoteMessage.getNotification().getBody();

    showNotification(title, message);
  }

  private void showNotification(String title, String message) {
```

```
        String channelId = "my_channel";

        NotificationManager manager =
                (NotificationManager) getSystemService(NOTIFICATION_SERVICE);

        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {

            NotificationChannel channel = new NotificationChannel(
                    channelId,
                    "Firebase Messages",
                    NotificationManager.IMPORTANCE_HIGH
            );

            manager.createNotificationChannel(channel);
        }

        NotificationCompat.Builder builder =
                new NotificationCompat.Builder(this, channelId)
                        .setSmallIcon(android.R.drawable.ic_dialog_info)
                        .setContentTitle(title)
                        .setContentText(message)
                        .setPriority(NotificationCompat.PRIORITY_HIGH)
                        .setAutoCancel(true);

        manager.notify(1, builder.build());
    }
}
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <uses-permission android:name="android.permission.POST_NOTIFICATIONS" />

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.SimpleDownstreamMessaging">

        <service
            android:name=".MyFirebaseMessagingService"
```

```
        android:exported="false">

        <intent-filter>
            <action android:name="com.google.firebase.MESSAGING_EVENT" />
        </intent-filter>

    </service>

    <activity
        android:name=".MainActivity"
        android:exported="true"
        android:windowSoftInputMode="adjustResize">

        <intent-filter>
            <action android:name="android.intent.action.MAIN" />

            <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>

    </activity>

  </application>

</manifest>
```